I. Stokes' theorem example

```
#showybox(
  title: "Stokes' theorem",
  frame: (
    border-color: blue,
    title-color: blue.lighten(30%),
    body-color: blue.lighten(95%),
    footer-color: blue.lighten(80%)
  ),
  footer: "Information extracted from a well-known public encyclopedia"
)[
  Let $Sigma$ be a smooth oriented surface in $RR^3$ with boundary $diff Sigma equiv Gamma$. If a vector field $bold(F)(x,y,z)=(F_x (x,y,z), F_y (x,y,z), F_z (x,y,z))$ is defined and has continuous first order partial derivatives in a region containing $Sigma$, then

  $ integral.double_Sigma (bold(nabla) times bold(F)) dot bold(Sigma) = integral.cont_(diff Sigma) bold(F) dot dif bold(Gamma) $
]
```

Stokes' theorem

Let Σ be a smooth oriented surface in $\mathbb{R}^3$ with boundary $\partial\Sigma \equiv \Gamma$. If a vector field $\boldsymbol{F}(x, y, z) = \left(F_x(x, y, z), F_y(x, y, z), F_z(x, y, z)\right)$ is defined and has continuous first order partial derivatives in a region containing Σ, then

$$\iint_{\Sigma} (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot \boldsymbol{\Sigma} = \oint_{\partial\Sigma} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{\Gamma}$$

Information extracted from a well-known public encyclopedia

II. Gauss's Law example

```
#showybox(
  frame: (
    border-color: red.darken(30%),
    title-color: red.darken(30%),
    radius: 0pt,
    thickness: 2pt,
    body-inset: 2em,
    dash: "densely-dash-dotted"
  ),
  title: "Gauss's Law"
)[
  The net electric flux through any hypothetical closed surface is equal to $1/epsilon_0$ times the net electric charge enclosed within that closed surface. The closed surface is also referred to as Gaussian surface. In its integral form:

  $ Phi_E = integral.surf_S bold(E) dot dif bold(A) = Q/epsilon_0 $
]
```

Gauss's Law

The net electric flux through any hypothetical closed surface is equal to $\frac{1}{\varepsilon_0}$ times the net electric charge enclosed within that closed surface. The closed surface is also referred to as Gaussian surface. In its integral form:

$$\Phi_E = \oiint_S \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A} = \frac{Q}{\varepsilon_0}$$

III. Carnot's cycle efficency example

```
#showybox(
  title-style: (
    weight: 900,
    color: red.darken(40%),
    sep-thickness: 0pt,
    align: center
  ),
  frame: (
    title-color: red.lighten(80%),
    border-color: red.darken(40%),
    thickness: (left: 1pt),
    radius: 0pt
  ),
  title: "Carnot cycle's efficency"
)[
  Inside a Carnot cycle, the efficiency $eta$ is defined to be:

  $ eta = W/Q_H = frac(Q_H + Q_C, Q_H) = 1 - T_C/T_H $
]
```

Carnot cycle's efficency

Inside a Carnot cycle, the efficiency η is defined to be:

$$\eta = \frac{W}{Q_H} = \frac{Q_H + Q_C}{Q_H} = 1 - \frac{T_C}{T_H}$$

IV. Clairaut's theorem example

```
#showybox(
  title-style: (
    boxed-style: (
      anchor: (
        x: center,
        y: horizon
      ),
      radius: (top-left: 10pt, bottom-right: 10pt, rest: 0pt),
```

```
    )
  ),
  frame: (
    title-color: green.darken(40%),
    body-color: green.lighten(80%),
    footer-color: green.lighten(60%),
    border-color: green.darken(60%),
    radius: (top-left: 10pt, bottom-right: 10pt, rest: 0pt)
  ),
  title: "Clairaut's theorem",
  footer: text(size: 10pt, weight: 600, emph("This will be useful every time you want to interchange partial derivatives in the future."))
)[
  Let $f: A arrow RR$ with $A subset RR^n$ an open set such that its cross derivatives of any order exist and are continuous in $A$. Then for any point $(a_1, a_2, ..., a_n) in A$ it is true that

  $ frac(diff^n f, diff x_i ... diff x_j)(a_1, a_2, ..., a_n) = frac(diff^n f, diff x_j ... diff x_i)(a_1, a_2, ..., a_n) $
]
```

Clairaut's theorem

Let $f : A \to \mathbb{R}$ with $A \subset \mathbb{R}^n$ an open set such that its cross derivatives of any order exist and are continuous in A. Then for any point $(a_1, a_2, ..., a_n) \in A$ it is true that

$$\frac{\partial^n f}{\partial x_i ... \partial x_j}(a_1, a_2, ..., a_n) = \frac{\partial^n f}{\partial x_j ... \partial x_i}(a_1, a_2, ..., a_n)$$

This will be useful every time you want to interchange partial derivatives in the future.

V. Divergence theorem example

```
#showybox(
  footer-style: (
    sep-thickness: 0pt,
    align: right,
    color: black
  ),
  title: "Divergence theorem",
  footer: [
    In the case of $n=3$, $V$ represents a volumne in three-dimensional space, and $diff V = S$ its surface
  ]
)[
  Suppose $V$ is a subset of $RR^n$ which is compact and has a piecewise smooth boundary $S$ (also indicated with $diff V = S$). If $bold(F)$ is a continuously differentiable vector field defined on a neighborhood of $V$, then:

```

```
  $ integral.triple_V (bold(nabla) dot bold(F)) dif V = integral.surf_S (bold(F) dot bold(hat(n))) dif S $
]
```

Divergence theorem

Suppose V is a subset of $\mathbb{R}^n$ which is compact and has a piecewise smooth boundary S (also indicated with $\partial V = S$). If $\boldsymbol{F}$ is a continuously differentiable vector field defined on a neighborhood of V, then:

$$\iiint_V (\boldsymbol{\nabla} \cdot \boldsymbol{F}) \, \mathrm{d}V = \oiint_S (\boldsymbol{F} \cdot \hat{\boldsymbol{n}}) \, \mathrm{d}S$$

In the case of $n = 3$, V represents a volumne in three-dimensional space, and $\partial V = S$ its surface

VI. Coulomb's law example

```
#showybox(
  shadow: (
    color: yellow.lighten(55%),
    offset: 3pt
  ),
  frame: (
    title-color: red.darken(30%),
    border-color: red.darken(30%),
    body-color: red.lighten(80%)
  ),
  title: "Coulomb's law"
)[
  Coulomb's law in vector form states that the electrostatic force $bold(F)$ experienced by a charge $q_1$ at position $bold(r)$ in the vecinity of another charge $q_2$ at position $bold(r')$, in a vacuum is equal to

  $ bold(F) = frac(q_1 q_2, 4 pi epsilon_0) frac(bold(r) - bold(r'), bar.v bold(r) - bold(r') bar.v^3) $
]
```

Coulomb's law

Coulomb's law in vector form states that the electrostatic force $\boldsymbol{F}$ experienced by a charge q_1 at position $\boldsymbol{r}$ in the vecinity of another charge q_2 at position $\boldsymbol{r}'$, in a vacuum is equal to

$$\boldsymbol{F} = \frac{q_1 q_2}{4\pi\varepsilon_0} \frac{\boldsymbol{r} - \boldsymbol{r}'}{\mid \boldsymbol{r} - \boldsymbol{r}' \mid^3}$$

VII. Newton's second law example

```
#block(
  height: 4.5cm,
  inset: 1em,
  fill: luma(250),
  stroke: luma(200),
  breakable: false,
  columns(2)[
    #showybox(
      title-style: (
        boxed-style: (
          anchor: (x: center, y: horizon)
        )
      ),
      breakable: true,
      width: 90%,
      align: center,
      title: "Newton's second law"
    )[
      If a body of mass $m$ experiments an acceleration $bold(a)$ due to a net force $sum bold(F)$, this acceleration is related to the mass and force by the following equation:

      $ bold(a) = frac(sum bold(F), m) $
    ]
  ]
)
```

Newton's second law

If a body of mass m experiments an acceleration $\boldsymbol{a}$ due to a net force $\sum \boldsymbol{F}$, this acceleration is related to the mass and force by the following equation:

$$\boldsymbol{a} = \frac{\sum \boldsymbol{F}}{m}$$

VIII. Encapsulation example

```
#showybox(
  title: "Parent container",
  lorem(10),
  columns(2)[
    #showybox(
      title-style: (boxed-style: (:)),
      title: "Child 1",
      lorem(10)
    )
    #colbreak()
    #showybox(
      title-style: (boxed-style: (:)),
```

```
      title: "Child 2",
      lorem(10)
    )
  ]
)
```

Parent container

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

Child 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

Child 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

IX. Breakable Styling example

```
#block(
  height: 10cm,
  inset: 1em,
  fill: luma(250),
  stroke: luma(200),
  breakable: false,
  columns(2)[
    #showybox(
      width: 100%,
      frame: (
        radius: 30pt,
        thickness: 2pt,
        border-color: blue,
        body-color: blue.lighten(80%),
        break-style: (
          upper-break: line(length: 100%, stroke: (dash: "dashed", paint: blue.darken(50%))),
          lower-break: line(length: 100%, stroke: (dash: "dashed", paint: blue.darken(50%))),
        ),
      ),
      breakable: true,
    )[
      #lorem(30)

      #lorem(40)

      #lorem(30)
    ]
  ],
)
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.